EXPERIMENT 3:

X-RAY DIFFRACTION

Date of Experiment : 05-08-2026 & 07-08-2026
Date of Submission : 20-08-2026

Akshara SomalaRaju
IMS24017

AIM:

To determine lattice parameters for single crystals and powders by analysing XRD data.

THEORY:

X-ray diffraction (XRD) is a versatile, non-destructive analytical technique used to determine phase composition, crystal structure, and crystallographic orientation in powder, solid, and liquid samples. Most crystalline materials consist of microscopic crystallites whose chemical composition and structure define their specific phase. Because each phase generates a unique diffraction pattern, identification can be performed by matching sample patterns against reference databases like the ICDD (International Centre for Diffraction Data). In multi-phase mixtures, the relative intensity of diffracted peaks enables quantitative determination of the full sample composition.

An X-ray diffractometer consists of three main components: an X-ray source, a sample holder, and a detector. Monochromatic X-rays illuminate the sample and undergo elastic scattering from parallel lattice planes separated by distance d. Constructive interference occurs when the path difference between scattered waves satisfies Bragg's Law:

$$n\lambda = 2d \sin \vartheta$$

where n is the diffraction order, λ is the X-ray wavelength, and ϑ is the glancing angle. By varying the angle between the incident and diffracted beams (2θ), the detector records scattered intensity. For cubic single crystals (such as NaCl or LiF), the inter-planar spacing for primary surface planes is $d = a_0/2$, simplifying Bragg's expression to

$$n\lambda = a_0 \sin \vartheta$$

which allows direct measurement of the lattice constant a_0.
In powder XRD (Debye - Scherrer method), randomly oriented crystallites diffract X-rays into concentric cones. Expressing the plane spacing in terms of Miller indices (h, k, l)yields the quadratic form:

$$\sin^2 \vartheta = \left(\frac{\lambda}{2a_0} \right)^2 \left[(nh)^2 + (nk)^2 + (nl)^2 \right] = F \cdot Z$$

In face - centred cubic structures like NaCl , phase interference between basis atoms restricts observed peaks to unmixed Miller indices (all-even or all-odd). All-even indices produce strong reflections, while all-odd reflections (e.g.,$\langle 111 \rangle$) are weaker due to atomic scattering factor differences ($f_A - f_H$). When the constituent ions are isoelectronic—such as K^+ and Cl^- in KCl — their scattering factors cancel out ($f_A \approx f_H$), causing all-odd reflections to disappear completely.

To study non-crystalline, amorphous, or nano structured components, complementary X-ray scattering techniques based on fundamental scattering theory are applied. Small-Angle X-ray Scattering (SAXS) and Grazing-Incidence SAXS (GISAXS) analyse nanoscale structural features and electron density variations. Pair Distribution Function (PDF) analysis uses total scattering data to examine short-range atomic ordering in disordered materials. Additionally, X-ray Reflectometry (XRR) measures specular reflection at small grazing angles to characterise thin-film thickness, density, and surface roughness.

PROCEDURE:

(i) Single Crystal XRD:

1. Mount the single crystal (e.g., NaCl) securely onto the target stage/ holder inside the bench top XRD unit and align the sample stage.
2. Set the target angle to 7.2° in the control software and wait briefly to verify and center the primary characteristic reflection line.
3. Configure a coupled scan in the software with a step width of $\Delta\beta = 0.1^\circ$ and a quick integration time of $\Delta t = 1$s per step. Run this scan for approximately 5 minutes to get a preliminary profile of the diffraction peaks.
4. Update the integration time to $\Delta t = 10$s per step while keeping $\Delta\beta = 0.1^\circ$. Execute this high-precision scan over the ~33-minute duration to capture full peak intensities with a high signal-to-noise ratio.
5. Analyse the intensity spectrum generated in the software after each scan run and record the precise peak positions (ϑ or 2θ values) for each diffraction order n.
6. Record the diffraction angle $\beta(^\circ)$ (2θ) and the corresponding X-ray intensity in arbitrary units (a.u.) in tabular form for both the survey scan ($\Delta t = 1$ s) and precision scan ($\Delta t = 10$s). From the recorded peak

angles, extract the glancing angles ϑ to calculate the lattice constant a_0 using Bragg's relation ($n\lambda = a_0 \sin\vartheta$)

(ii) Powdered XRD:

1. Grind the salt thoroughly using a mortar and pestle for approximately 10 minutes to achieve a fine, homogeneous powder with randomly oriented crystallites.
2. Load the finely pulverised salt into the XRD sample holder, pressing and levelling the powder surface to ensure uniform sample density.
3. Place the prepared sample holder securely onto the target stage inside the XRD unit and set up the 2θ coupled scan mode.
4. Configure the scanning parameters in the software with a step width of $\Delta\beta = 0.1^\circ$ and an integration time of $\Delta t = 10$s per step.
5. Execute the automated scan across the angular range and let it run for approximately 50 minutes to record high-resolution diffraction peaks.
6. Record the peak positions ($\beta(^\circ)(2\theta)$) and their corresponding diffracted intensities (a.u.) in tabular form from the resultant diffraction spectrum.
7. Compute the $\sin^2\vartheta$ values for each peak, assign Miller indices (h, k, l) according to selection rules for cubic structures (e.g., unmixed indices for fcc), and calculate the inter-planar spacing d and lattice constant a_0.

OBSERVATIONS:

(i) Single Crystal XRD:

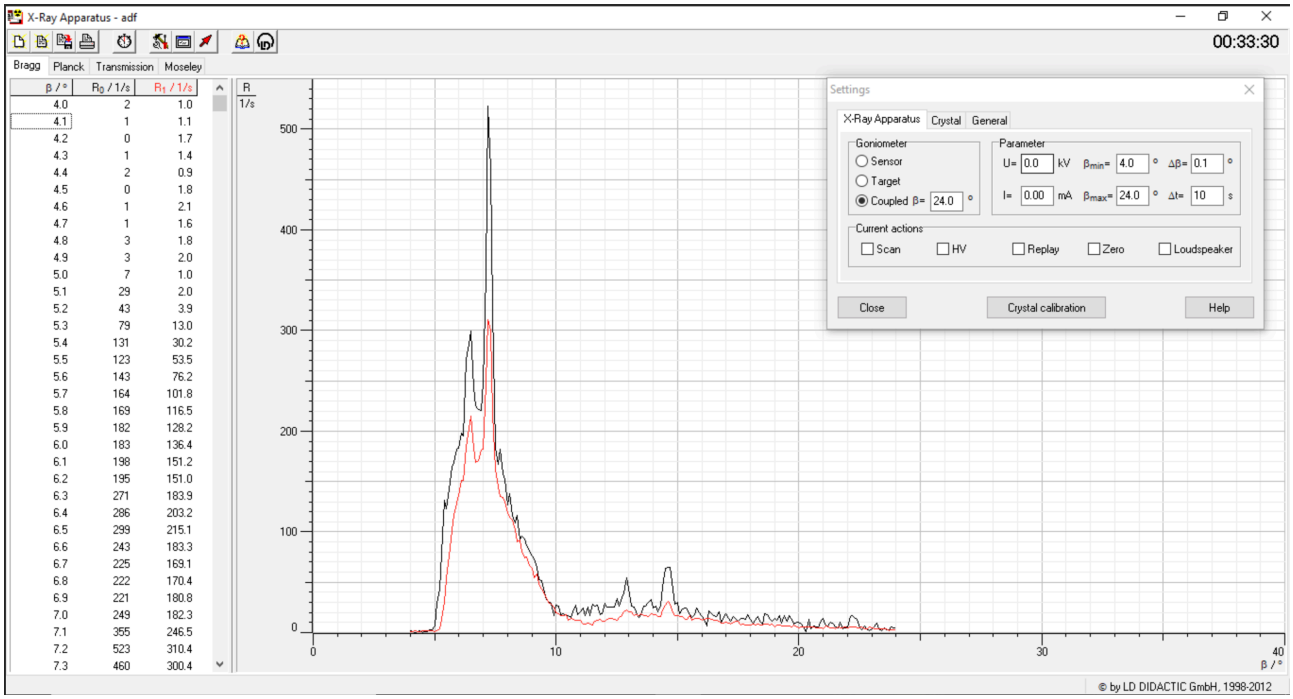


Table 1:

($\Delta t = 1s$)

R (1/s)	$\beta(\theta)(^\circ)$	λ (pm)	n	sin(θ)	nλ
299	6.5	63.09	1	0.1132	63.09
523	7.2	71.08	1	0.1253	71.08
54	12.9	63.09	2	0.2232	126.18
65	14.7	71.08	2	0.2537	142.16

Graph 1:

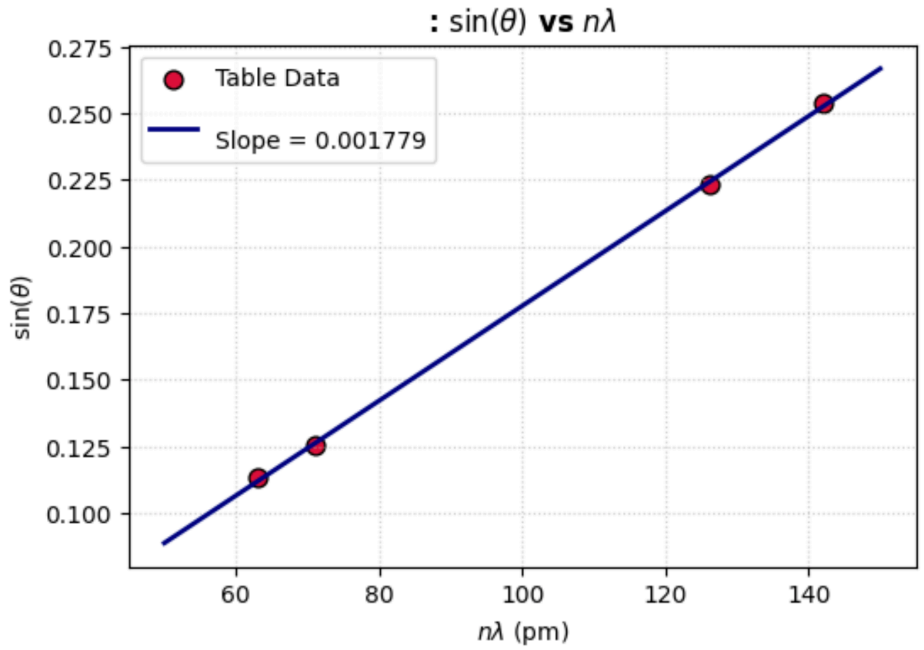
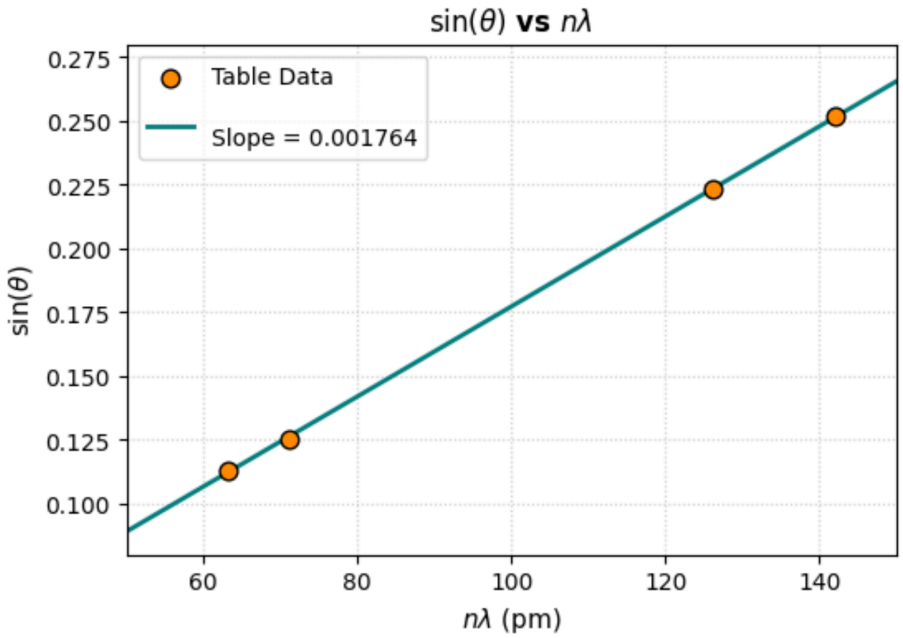


Table 2:

($\Delta t = 10s$)

R (1/s)	$\beta(\theta)(^\circ)$	λ (pm)	n	sin(θ)	nλ
215.1	6.5	63.09	1	0.1132	63.09
310.4	7.2	71.08	1	0.1253	71.08
22.5	12.9	63.09	2	0.2232	126.18
30.8	14.6	71.08	2	0.2520	142.16

Graph 2:



(ii) Powdered XRD:

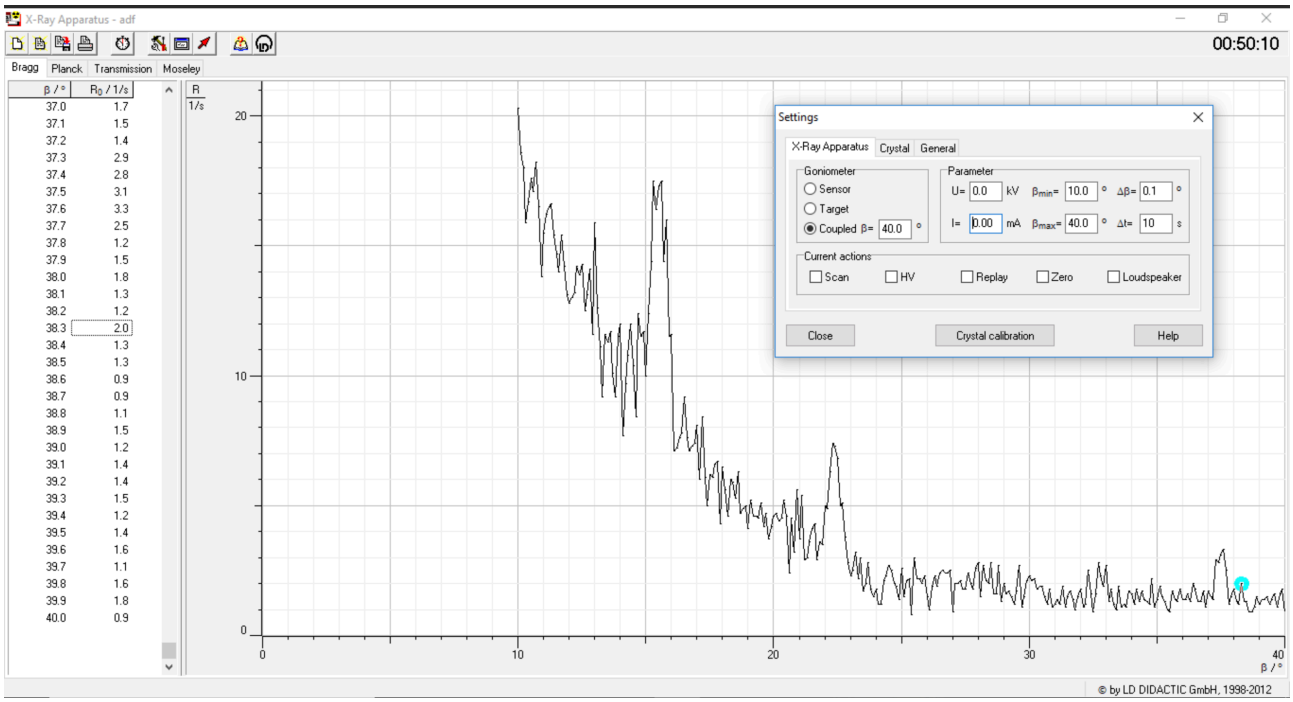


Table 3:

R (1/s)	$\beta(\theta)(^\circ)$	Miller's Indices (hkl)	n	$((nh)^2+(nk)^2+(nl)^2)$	sin(θ)	a_0(pm)
17.5	15.6	200	1	4	0.2689	573.38
7.4	22.3	220	1	8	0.3794	574.72
2.7	29.6	222	1	12	0.4939	540.70
2.8	32.7	400	1	16	0.5402	570.84
3.3	37.6	420	1	20	0.6101	565.09

CALCULATIONS AND ANALYSIS:

(i) Single Crystal XRD:

According to Bragg's law:

$$n\lambda = 2d\sin(\theta)$$

Rearranging the equation to isolate $\sin(\theta)$ on the vertical axis gives:

$$\sin(\theta) = \left(\frac{1}{2d}\right) n\lambda$$

For an NaCl crystal structure, $d = \frac{a_0}{2}$. Substituting this into the slope formula yields:

$$m = \frac{1}{2\left(\frac{a_0}{2}\right)} = \frac{1}{a_0}$$

Solving explicitly for the lattice constant a_0 with our experimental slope $m1 = 0.001764$ pm^{-1}:

$$a_1 = \frac{1}{m_1}$$

$$a_1 = \frac{1}{0.001764 \text{ pm}^{-1}}$$

$$a_1 \approx \mathbf{566.9} \text{ pm}$$

$m2 = 0.001779$ pm^{-1}:

$$a_2 = \frac{1}{m_2}$$

$$a_2 = \frac{1}{0.001779\ \text{pm}^{-1}}$$

$$a_2 \approx \mathbf{562.1}\ \text{pm}$$

$$a_{mean} = (a_1 + a_2)/2$$

$$a_{mean} = 564.5pm$$

Error Calculation:

$$\text{Percentage Error} = \left| \frac{\text{Experimental Value} - \text{Theoretical Value}}{\text{Theoretical Value}} \right| \times 100\,\%$$

$a_{\text{exp}} = 564.5\ \text{pm}\ and\ a_{\text{theo}} = 564\ \text{pm}$

$$\text{Percentage Error} = \left| \frac{564.5 - 564}{564} \right| \times 100\,\%$$

$$= \left| \frac{0.5}{564} \right| \times 100\,\%$$

$$\approx \mathbf{0.0886}\,\%$$

(ii) Powdered XRD:

$$a_0 = \frac{\lambda}{2\sin(\theta)} \sqrt{(nh)^2 + (nk)^2 + (nl)^2}$$

1. $$a_1 = \frac{154.184}{2 \times 0.2689} \times 2$$

$$a_1 = 573.38\ \text{pm}$$

2. $$a_2 = \frac{154.184}{2 \times 0.3794} \times \sqrt{8}$$
$$a_2 = 574.72 \text{ pm}$$

3. $$a_3 = \frac{154.184}{2 \times 0.4939} \times \sqrt{12}$$
$$a_3 = 540.70 \text{ pm}$$

4. $$a_4 = \frac{154.184}{2 \times 0.5402} \times 4$$
$$a_4 = 570.84 \text{ pm}$$

5. $$a_5 = \frac{154.184}{2 \times 0.6101} \times \sqrt{20}$$
$$a_5 = 565.09 \text{ pm}$$

$$a_{mean} = (a_1 + a_2 + a_3 + a_4 + a_5)/5$$

$$a_{mean} = 564.95pm$$

Error Calculation:

$$a_{\text{exp}} = 564.95 \text{ pm } and \ a_{\text{theo}} = 564 \text{ pm}$$

$$\text{Percentage Error} = \left| \frac{564.95 - 564}{564} \right| \times 100\,\%$$

$$= \left| \frac{0.95}{564} \right| \times 100\,\%$$

$$\approx \mathbf{0.168}\,\%$$

RESULT:

(i) In the mono crystal X-ray diffraction from the linear regression analysis of the inverted Bragg's Law plot ($\sin\theta$ vs. $n\lambda$), the lattice constant was determined directly from the slope as $a_0 = 564.50$ pm

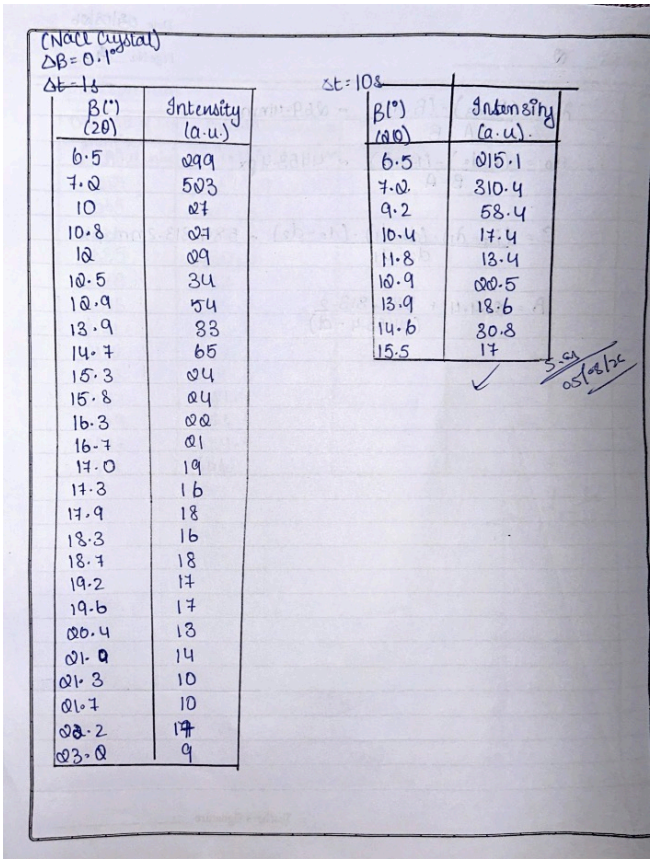
(ii) In the powdered X-ray diffraction by calculating the lattice constant individual reflection peaks indexed from the powder diffraction pattern, the statistical average value was determined to be $a_{\text{mean}} = 564.95$ pm

DISCUSSION AND REMARKS:

The experimental analysis demonstrates the distinct operational strengths and physics of determining crystal lattice parameters using both single-crystal (mono-crystal) and powdered (polycrystal) X-ray diffraction (XRD) techniques. The mono crystal analysis highlights the analytical advantage of isolating a singular, uninterrupted crystal lattice orientation. Because a single-crystal sample aligns continuously in three dimensions, its constructive interference reflections generate sharp, highly defined peaks with minimal background noise. This high signal-to-noise ratio allows for an exceptionally reliable linear regression analysis when mapping the experimental variables against one another. The calculated slope directly reflects the true face-centred cubic (FCC) unit cell dimensions, confirming that single-crystal diffraction remains the definitive benchmark method for capturing undisturbed structural parameters.

In contrast, the polycrystal method reveals the unique profile characteristics that arise from bulk, multi-oriented materials. Milling the substance into a fine powder crushes the unified crystal lattice into millions of tiny, randomly oriented micro-crystallites. Consequently, at any specific Bragg angle, only a tiny fraction of these grains are oriented precisely enough to satisfy the conditions for constructive interference. This random spatial distribution accounts for the significantly lower peak intensities and broader, more diffuse profiles observed on the powder spectrum compared to the sharp spikes of the single-crystal data. Despite these structural variances and minor instrumental broadening effects, calculating the statistical mean across multiple distinct index planes—such as the ((200), (220), (222), (400), and (420)) orientations—successfully minimises individual data anomalies. Ultimately, both techniques prove highly effective at validating the underlying cubic symmetry and crystal structural identity of the sample.

LAB RECORD:



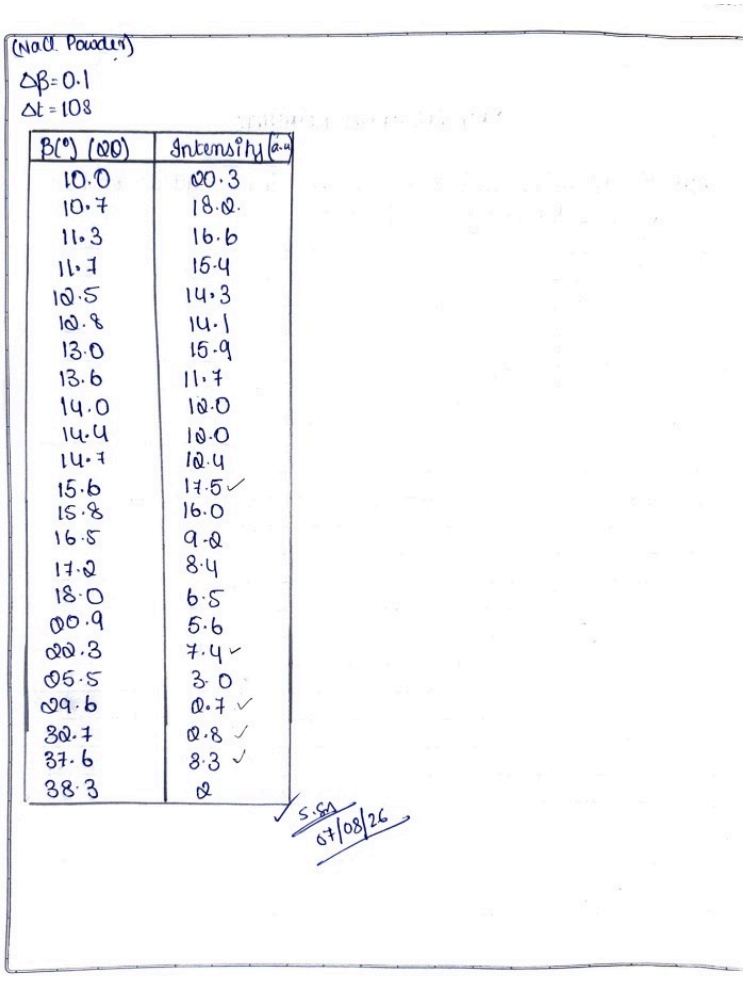
(NaCl Crystal)
$\Delta\beta = 0.1^{\circ}$
$\Delta t = 1s$

$\beta(^{\circ})$ (2θ)	Intensity (a.u.)
6.5	0099
7.0	503
10	07
10.8	07
10	09
10.5	34
10.9	54
13.9	83
14.7	65
15.3	04
15.8	04
16.3	002
16.7	01
17.0	19
17.3	16
17.9	18
18.3	16
18.7	18
19.2	17
19.6	17
20.4	13
21.0	14
21.3	10
21.7	10
22.2	17
23.0	9

$\Delta t = 10s$

$\beta(^{\circ})$ (100)	Intensity (a.u.)
6.5	0115.1
7.0	310.4
9.2	58.4
10.4	17.4
11.8	13.4
10.9	000.5
13.9	18.6
14.6	30.8
15.5	17

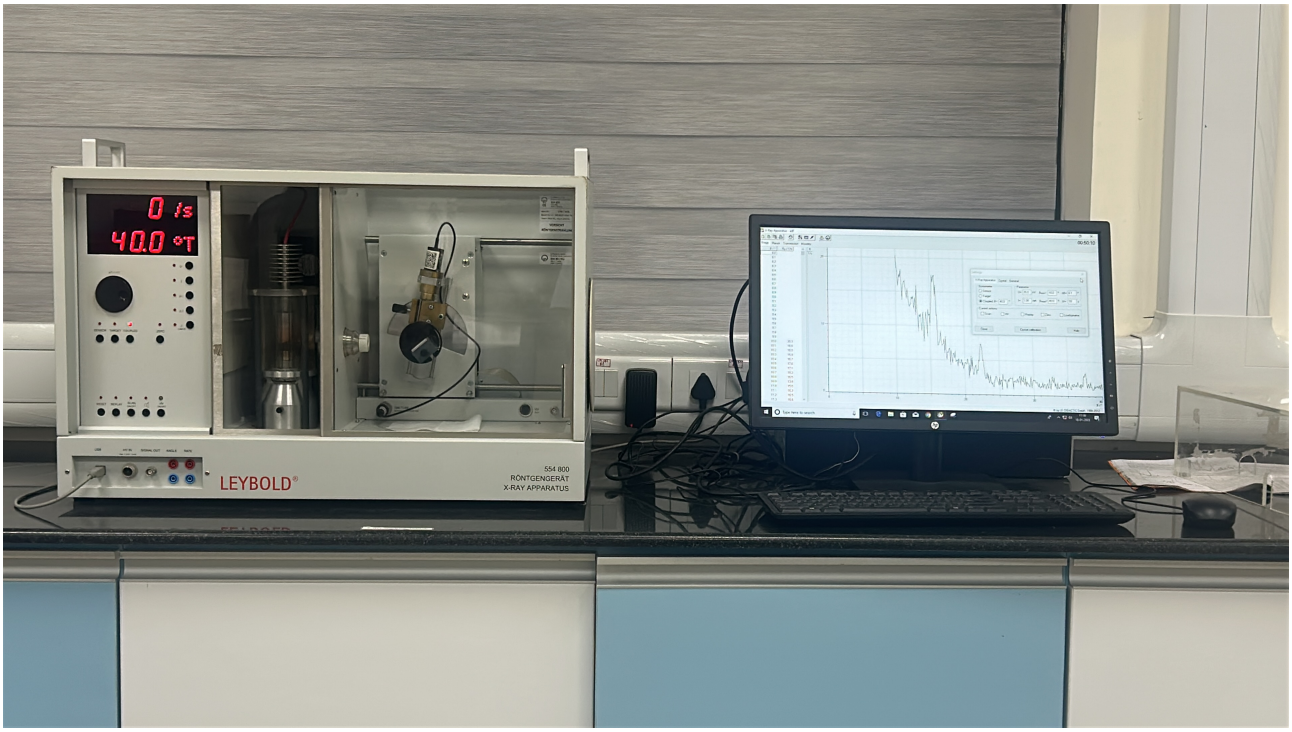
S.SA 05/08/26



(NaCl Powder)
$\Delta\beta = 0.1$
$\Delta t = 10s$

$\beta(^{\circ})$ (00)	Intensity (a.u)
10.0	00.3
10.7	18.0
11.3	16.6
11.7	15.4
10.5	14.3
10.8	14.1
13.0	15.9
13.6	11.7
14.0	10.0
14.4	10.0
14.7	10.4
15.6	17.5 ✓
15.8	16.0
16.5	9.0
17.2	8.4
18.0	6.5
00.9	5.6
00.3	7.4 ✓
05.5	3.0
09.6	0.7 ✓
30.7	0.8 ✓
37.6	8.3 ✓
38.3	0

S.SA 07/08/26



REFERENCES:

- https://www.malvernpanalytical.com/en/products/technology/xray-analysis/x-ray-diffraction
- LEYBOLD Physics Leaflets : Solid-state physics Properties of crystals X-ray structural analysis